

Similarity-Weighted Linguistic Time Series Forecasting using Hedge Algebras Semantic Kernels

Nguyen Duy Hieu

Faculty of Natural Sciences, Tay Bac University, Son La, Viet Nam
hieund@utb.edu.vn

Abstract. Linguistic Time Series (LTS) forecasting, grounded in Hedge Algebras (HA), provides an interpretable algebraic approach to short-sequence prediction. However, all existing LTS models rely on *exact-match lookup*, producing a forecast only when the current λ -gram appears verbatim in the rule base—an all-or-nothing inference that discards the graded semantic structure of HA. We formally characterize this *sparse-rule problem* via coverage ratio $|\mathcal{R}|/|W|^\lambda$, which reaches 0.56% for Alabama at $\lambda = 3$. We propose **SW-LTS** (Similarity-Weighted LTS): an interpretable graded inference mechanism in which every prediction is a linguistically auditable weighted average of stored rules, with weights derived from Gaussian kernels over normalized HA semantic distances. Any forecast can be explained by ranking rules by their kernel weight—semantic proximity in the algebraic structure of HA, not geometric distance in raw data space. Two formal propositions establish strict generalization over LTS and HO-LTS. Empirical results confirm SW-LTS matches or improves in-sample accuracy and achieves a statistically significant 69.5% walk-forward MSE reduction on Wolf’s Sunspot ($p < 0.001$), with interpretability demonstrated via kernel weight attribution on a case study (Table 5).

Keywords: fuzzy time series, hedge algebras, linguistic time series, interpretable forecasting, kernel-weighted aggregation, sparse-rule problem

1 Introduction

Fuzzy Time Series (FTS), introduced by Song and Chissom [13, 14], encodes observations as linguistic labels and extracts forecasting rules from sequential patterns, offering interpretability where deep-learning models require much larger corpora [16]. Linguistic Time Series (LTS) [4] advances FTS by replacing ad hoc interval partitioning with Hedge Algebras (HA) [8, 9]: a formal algebraic structure that equips each term with an analytically computable semantic value. Subsequent extensions include high-order LTS (HO-LTS [5]), PSO-tuned optimization (LTS-PSO [6]), and co-optimization (CO-LTS [7]).

Despite these advances, all LTS models share one limitation: **exact-match lookup**—a forecast is produced only when the current λ -gram appears verbatim in the rule base. The resulting **sparse-rule problem** ($|\mathcal{R}|/|W|^\lambda$ collapses exponentially) drives coverage to 0.56% for Alabama at $\lambda = 3$. HO-LTS replaces the fallback but retains binary lookup; LTS-PSO and CO-LTS tune parameters without fixing the lookup mechanism.

We propose **SW-LTS** (Similarity-Weighted LTS), replacing discrete lookup with Gaussian kernel-weighted aggregation in normalized HA semantic space (Nadaraya–Watson estimator [12]). Every prediction is a linguistically graded weighted average of stored rules with weights ranked by HA semantic proximity, preserving and extending LTS interpretability. Bandwidth σ , dimensionless in $[0, 1]^\lambda$, is the sole additional parameter.

2 Background

2.1 Hedge Algebras and Semantic Quantifying Measure

A Hedge Algebra [8] is a quintuple $\mathcal{AX} = (X, G, C, H, \leq)$, where X is the universe of linguistic terms, $G = \{c^-, c^+\}$ are generators (*Small* and *Large*), $C = \{w_0\}$ is the neutral element, H is a set of hedges, and $\leq$ is a semantically consistent linear order. Two primary hedges V (*Very*) and L (*Rather*) are applied recursively: at specificity k , the vocabulary contains $|W| = 2^{k+2} - 1$ terms ($k = 2$: 15; $k = 3$: 31 terms).

The **Semantic Quantifying Measure** (SQM) [9] $\nu : X \rightarrow [0, 1]$ assigns each term a unique value via parameters $\theta = \nu(c^-)$ and α (weight of L). For universe $[\ell, u]$, the **semantic point** of w is

$$s(w) = \ell + (u - \ell) \nu(w) \in [\ell, u]. \quad (1)$$

2.2 LTS Model and Related Context

LTS [4] encodes each observation as $\ell_t = \arg \min_w |s(w) - y_t|$, builds order- λ Linguistic Logical Relationship Groups (LLRG) $\mathcal{R} : \mathbf{k} \mapsto R$ from training λ -grams, and forecasts by

$$\hat{y}_t = \begin{cases} \frac{1}{|R|} \sum_{r \in R} s(r) & \text{if } \ell_t \in R, \\ s(\ell_{t-1}) & \text{(no-match fallback).} \end{cases} \quad (2)$$

Three structural limitations arise: **L1 (Sparsity)**—coverage decays exponentially; **L2 (Discontinuity)**—one-hedge-apart patterns yield completely different forecasts; **L3 (Binary confidence)**—matched and unmatched queries are treated identically. HO-LTS replaces the fallback but retains binary lookup; LTS-PSO and CO-LTS tune (θ, α) without fixing it.

The idea of smoothing across sparse rule sets has precedents in fuzzy rule interpolation: Kóczy and Hirota [11] showed missing rules can be approximated

by linear interpolation in consequent space. SW-LTS differs by operating within the HA algebraic structure, where distances between antecedent patterns are grounded in algebraically defined semantic values. Yu [15] weighted FTS rules by temporal recency rather than semantic similarity, which does not address structural sparsity. Recent FTS advances focus on non-stationary partitioning [1] and interpretable rule attribution [2]; SW-LTS is the first to ground weighted rule inference in the algebraic structure of Hedge Algebras, making pairwise distances axiomatically defined rather than geometrically arbitrary.

3 The Proposed SW-LTS Model

3.1 Semantic Representation

The **query window** at time t is the λ -gram $\ell_t = (\ell_{t-\lambda}, \dots, \ell_{t-1}) \in W^\lambda$ (bold notation; plain ℓ_t denotes a single label as in Section 2.2).

Definition 1 (Normalized LHS Vector). For $\mathbf{w} = (w_1, \dots, w_\lambda) \in W^\lambda$, the normalized semantic vector is $\tilde{\phi}(\mathbf{w}) = \frac{1}{\Delta}(s(w_1) - \ell, \dots, s(w_\lambda) - \ell) \in [0, 1]^\lambda$, where $\Delta = u - \ell$.

Normalization maps all LHS patterns into $[0, 1]^\lambda$ irrespective of data scale, making pairwise distances dimensionless.

3.2 Gaussian Kernel and SW-LTS Forecast

We define pattern similarity via a Gaussian kernel on normalized vectors:

$$K(\mathbf{u}, \mathbf{v}) = \exp\left(-\frac{\|\tilde{\phi}(\mathbf{u}) - \tilde{\phi}(\mathbf{v})\|^2}{2\sigma^2}\right), \quad \sigma > 0. \quad (3)$$

Let $\mathcal{R} = \{(\mathbf{k}_j, R_j)\}_{j=1}^M$ be the LLRG and $\bar{s}(R_j) = \frac{1}{|R_j|} \sum_{r \in R_j} s(r)$ the mean RHS semantic point. The SW-LTS forecast applies the Nadaraya–Watson estimator [12] in HA semantic space:

$$\hat{y}_t^{\text{SW}} = \frac{\sum_{j=1}^M K(\ell_t, \mathbf{k}_j) \bar{s}(R_j)}{\sum_{j=1}^M K(\ell_t, \mathbf{k}_j)}, \quad (4)$$

with fallback $\hat{y}_t = s(\ell_{t-1})$ when the denominator is zero. Every query uses *all* M rules with semantically graded weights.

Interpretability and complexity. The normalized weight $w_j = K(\ell_t, \mathbf{k}_j) / \sum_i K(\ell_t, \mathbf{k}_i)$ quantifies the *linguistic relevance* of rule j : semantically adjacent patterns receive high weight, distant patterns low weight. Any forecast is auditable by ranking rules by w_j —an explanation grounded in the HA algebraic structure, not in raw numeric distance. Per-query cost is $O(M\lambda)$ with $M \leq 139$, comparable to HO-LTS.

3.3 Theoretical Properties

Proposition 1 ($\sigma \rightarrow 0$: Recovery of LTS). *As $\sigma \rightarrow 0$, kernel mass concentrates on the exact-match rule if it exists, and (4) reduces to (2); with no match the fallback $s(\ell_{t-1})$ applies. SW-LTS exactly recovers LTS.*

Proposition 2 ($\sigma \rightarrow \infty$: Recovery of Global-Mean Fallback). *As $\sigma \rightarrow \infty$, all kernel weights equal 1, and (4) reduces to $\frac{1}{M} \sum_j \bar{s}(R_j)$ —the global-mean fallback of HO-LTS.*

Both results follow from the exponential form of K , establishing SW-LTS as a *strict generalization*: any $\sigma \in (0, \infty)$ interpolates between exact-match and global-mean extremes.

4 Experiments

4.1 Experimental Setup

Datasets (Table 1). Alabama Enrollment ($n = 22$, [13]) is the primary sparse-coverage benchmark. TAIEX 1999 ($n = 241$, [15]), Wolf’s Sunspot ($n = 288$), and Tuscaloosa Temperature ($n = 122$) cover finance, astronomy, and meteorology domains.

Table 1. Benchmark Datasets

Dataset	n	Universe $[\ell, u]$	Domain
Alabama Enrollment	22	[13,000, 20,000]	Education
TAIEX 1999	241	[5,201, 9,039]	Finance
Wolf’s Sunspot	288	[0, 296.23]	Astronomy
Tuscaloosa Temp.	122	[23, 32]	Meteorology

Baselines: Song & Chissom [13], Chen [3], LTS [4], HO-LTS [5] ($\lambda \in \{2, 3\}$), and LTS-PSO [6]. All LTS-family models use $k = 2$, $\theta = 0.57$, $\alpha = 0.49$.

Evaluation. (i) *In-sample* follows the standard LTS protocol [4, 5] for direct baseline comparison. (ii) *Walk-forward* uses an expanding window (initial fold: 70% Alabama, 80% others); σ^* is fixed from the initial fold then the model retrains up to each step t , predicts y_t , and results are compared against a naive last-value predictor $\hat{y}_t = y_{t-1}$. Bootstrap 95% CI (10,000 resamples) quantifies uncertainty in paired MSE differences.

4.2 In-Sample Results

Table 2 reports MSE and MAE. SW-LTS ($\lambda = 3$) achieves the lowest MSE on Alabama (14,420 vs. 15,077 for HO-LTS: **4.4%**) and matches HO-LTS on TAIEX-99, Sunspot, and Temperature—where high coverage drives SW-LTS

to the exact-match regime ($\sigma^* = 0.01$, Proposition 1). At $\lambda = 2$, SW-LTS outperforms HO-LTS on TAIEX-99: MSE 18,283 vs. 20,255 (**9.7%**, coverage 20%, 80% of queries unmatched).

Table 2. In-sample MSE and MAE (bold = best; underline = 2nd best; † SW-LTS $\sigma^* \rightarrow 0$ recovers exact-match, tied with HO-LTS by Proposition 1)

Method	Alabama		TAIEX-99		Sunspot		Temp.	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Song & Chissom (1993)	806,087	632.1	103,899	239.9	2,584	35.5	3.77	1.66
Chen (1996)	407,521	498.8	42,617	172.0	3,342	51.7	1.08	0.86
LTS (2020)	123,525	265.7	26,517	129.7	2,363	42.7	0.79	0.74
HO-LTS, $\lambda = 2$	64,082	192.0	20,255	107.9	907	22.3	0.44	0.51
HO-LTS, $\lambda = 3$	<u>15,077</u>	<u>112.4</u>	14,728	93.1	562	16.2	0.16	0.25
LTS-PSO (2022)	140,525	307.3	15,477	97.2	1,382	31.0	0.77	0.74
SW-LTS, $\lambda = 2$	64,082	192.0	<u>18,283</u>	<u>102.4</u>	907	22.3	0.44	0.51
SW-LTS, $\lambda = 3$	14,420	107.8	14,728†	93.1†	562†	16.2†	0.16†	0.25†

4.3 Out-of-Sample Walk-Forward Validation

Table 3 reports walk-forward MSE at $\lambda = 3$. On Sunspot, SW-LTS (MSE 1,100) achieves **69.5%** reduction vs. HO-LTS (3,604) and beats the naive predictor (2,155), $p < 0.001$. Temperature reverses: SW-LTS (2.145) is 47.7% worse than HO-LTS (1.452), both exceeding naive (0.891). Alabama ($n_{\text{test}}=7$) is included for completeness only.

Table 3. Walk-forward evaluation ($\lambda = 3$). Naive: $\hat{y}_t = y_{t-1}$. $\Delta(\%)$: SW-LTS vs. HO-LTS (from unrounded values; Temperature shown to 3 d.p.). p : one-sided bootstrap $P[\text{MSE}(\text{HO}) - \text{MSE}(\text{SW}) \leq 0]$. *** $p < 0.001$; $^\dagger n_{\text{test}}=7$, completeness only.

Dataset	n_{test}	Naive MSE	HO-LTS MSE	SW-LTS MSE	$\Delta(\%)$	p
Alabama †	7	588,502	1,915,359	1,191,436	-37.8	0.121
TAIEX-99	49	10,995	22,781	22,909	+0.6	0.497
Sunspot	58	2,155	3,604	1,100	-69.5	<0.001***
Temp.	25	0.891	1.452	2.145	+47.7	0.969

4.4 Coverage and Bandwidth Analysis

Table 4 reports LLRG coverage at $\lambda = 1, 2, 3$; Figure 1(a) visualizes the decay profile. Alabama is most extreme: 73.3% at $\lambda = 1$ collapses to 0.56% at $\lambda = 3$ (19 of $15^3 = 3,375$ possible). SW-LTS assigns non-zero weight to all 19 rules for every query. Coverage explains the in-sample pattern: on TAIEX-99 at $\lambda = 2$ (coverage 20%), the 9.7% gain arises because 80% of queries no longer fall to the global mean.

Table 4. LLRG coverage (%) at $k = 2$, $|W| = 15$; $|\mathcal{R}|$ in parentheses

Dataset	$\lambda = 1$	$\lambda = 2$	$\lambda = 3$
Alabama	73.33% (11)	7.56% (17)	0.56% (19)
TAIEX-99	93.33% (14)	20.00% (45)	2.64% (89)
Sunspot	86.67% (13)	31.56% (71)	4.12% (139)
Temp.	100.00% (15)	26.22% (59)	3.11% (105)

Figure 1(b) plots MSE vs. σ on Alabama ($\lambda = 3$). For $\sigma \leq 0.01$, MSE is flat at 15,077 (exact-match regime, Proposition 1); at $\sigma^* = 0.03$ MSE reaches 14,420; for $\sigma > 0.03$ MSE rises toward the global-mean limit (Proposition 2).

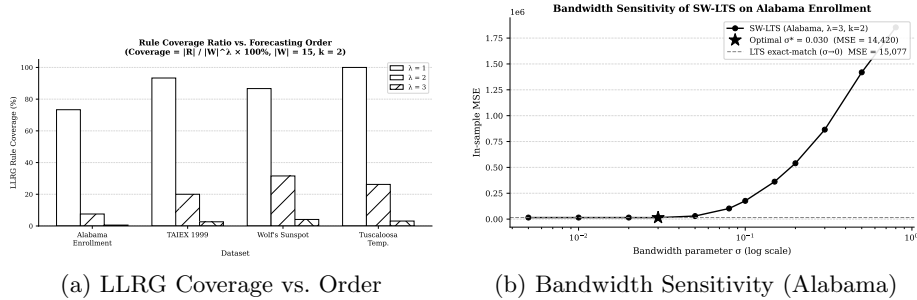


Fig. 1. (a) Coverage (%) at $\lambda = 1, 2, 3$ for all datasets. (b) MSE vs. σ on Alabama ($\lambda = 3$). Star: in-sample optimum $\sigma^* = 0.03$ (MSE = 14,420); dashed: LTS exact-match baseline (MSE = 15,077).

4.5 Discussion

Sunspot: kernel aggregation over high-dynamic-range cycles. Sunspot numbers span 0–296, so HO-LTS’s global-mean fallback for $\approx 96\%$ of unmatched queries is far from the actual value in both active and quiet solar phases. SW-LTS ($\sigma^* = 0.020$) kernels over all 139 training rules, weighting semantically adja-

cent patterns, and achieves $\text{MSE} = 1,100$ —below both HO-LTS (3,604, -69.5% , $p < 0.001$) and the naive predictor (2,155).

Temperature: limits of kernel smoothing. On Tuscaloosa Temperature (23–32°C), SW-LTS walk-forward MSE is 47.7% higher than HO-LTS (2.145 vs. 1.452, $p = 0.969$), with both exceeding naive (0.891). The narrow range makes the global mean a reliable fallback; kernel-weighted rules from phase-shifted seasons introduce additional error. SW-LTS’s benefit requires high dynamic range with structured periodicity.

Interpretability: kernel weight attribution. Table 5 decomposes walk-forward step $t=237$ on Sunspot ($y_t=190.6$, $\hat{y}_t=190.1$, error 0.3%). Rule 1 (exact match, Prop. 1) takes 99.5% weight; Rules 2–3, at HA distance ≈ 0.07 , share 0.3%; all 110 remaining rules together contribute $< 0.2\%$. This attribution is impossible under binary lookup and is auditable in the HA algebraic structure.

Table 5. Kernel weight attribution, $t=237$, Wolf’s Sunspot ($y_t=190.6$, $\hat{y}_t=190.1$; $\sigma^*=0.020$, $\lambda=3$, $M=113$). Query: $\ell_t=(VV^-, LV^-, L^-)$ ($-$ =hedge on c^- , Small; e.g., $VV^- = V(V(c^-))$).

Rank	LHS k_j	$\bar{s}(R_j)$	w_j
1	(VV^-, LV^-, L^-)	190.1	99.5%
2	(VV^-, LV^-, LL^-)	147.2	0.2%
3	(V^-, LV^-, L^-)	199.4	0.1%
Remaining 110 rules			$< 0.2\%$

Why LTS-PSO cannot fix sparsity. With $n-\lambda=19$ λ -grams, the 0.56% ceiling holds regardless of (θ, α) : the lookup mechanism must change.

5 Conclusion

SW-LTS advances LTS forecasting on two fronts. *Interpretability*: graded kernel aggregation makes every prediction a linguistically auditable weighted average of stored rules, with weights derived from the HA algebraic structure—demonstrated via kernel weight attribution at $t=237$ on Sunspot (Table 5). *Generalization*: we formally characterize the sparse-rule problem (coverage decays to 0.56% on Alabama at $\lambda=3$) and prove SW-LTS recovers both LTS ($\sigma \rightarrow 0$) and HO-LTS ($\sigma \rightarrow \infty$). In-sample experiments confirm SW-LTS matches or improves the best baseline across four datasets; walk-forward validation shows a significant 69.5% MSE reduction on Wolf’s Sunspot ($p < 0.001$). Future work includes adaptive bandwidth, multi-step forecasting, and non-Gaussian kernels compatible with the HA partial order.

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